

Algebra II - Tutorial 8 - Date: 24/3/2025

1. Prove that a group G is free on a subset X if and only if X generates G and no reduced word in $X \cup X^{-1}$ of positive length is the identity. ($\ell(w) :=$ length of the word is the length of a reduced word)

Hint:—Show that the inclusion map $X \rightarrow G$ extends to a unique isomorphism $F(X) \rightarrow G$

2. Show that free groups $F(X)$ do not have elements of finite order > 1 . In fact, if $a^n = b^n$ for a, b in a free group F , prove that $a = b$.

Hint: Need to consider words of the form $w = w_1^{-1}w_2w_1$ and consider w^n

3. Let F be a free group let $w \in F$ Show that $\{a \in F : a^n = w\}$ is finite

Hint: Consider $\ell(w)$ and $\ell(a^n)$

4. Show that in a free group, two commuting elements a, b must satisfy $a = c^u$ and $b = c^v$ for some element c and some integers u, v . In particular, a free group has nontrivial center if and only if its rank is 1.
5. If a, b are elements in a free group F , and satisfy $a^p b^q = b^q a^p$ for some non-zero integers p, q then prove a, b are integer powers of a common element.
6. If $w \neq 1$ in a free group F , then prove that the centralizer $C(w)$ is an infinite cyclic group.
7. If $w \neq 1$ in a free group, show that w cannot be conjugate to w^{-1} .
8. If N is a normal subgroup of a group G such that G/N is free, then prove that there exists a subgroup H of G satisfying $G = HN$ and $H \cap N = \{1\}$.
9. Let H be a subgroup of infinite index in a free group F . Show that for each subgroup $K \neq \{1\}$ of F , $H \cap K \neq \{1\}$.